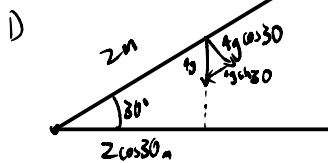


Moments 2

Consolidation



$$4T = 2 \times 4g \cos 30$$

$$T = g\sqrt{3}$$

$$= 16.97 \text{ N}$$

$$\sqrt{(16.97 - 4g \cos 30)^2 + (4g \sin 30)^2} = \text{magnitude}$$

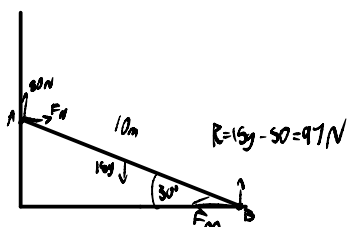
$$= 25.9$$

$$\theta = \arctan\left(\frac{16.97 - 4g \cos 30}{4g \sin 30}\right) + 30^\circ$$

$$= 40.9^\circ + 30^\circ$$

$$= 70.9^\circ$$

2



$$i) 50 \times 10 \cos 30 + 10 \sin 30 \times F_N = 5 \cos 30 \times 15g$$

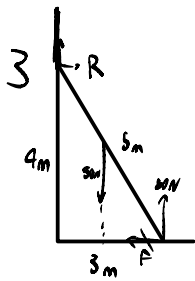
$$F_N = 40.7$$

$$ii) 9.7 \text{ N}$$

$$iii) F_N = \text{frictional force};$$

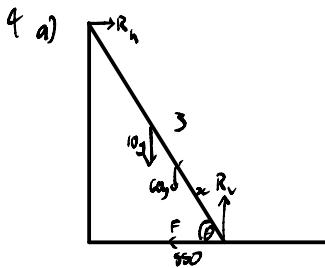
particles in equilibrium

$$\therefore \text{horizontal forces sum to } 0$$



$$50 \times 1.5 = R \times 4$$

$$R = \frac{75}{4} = 18.75 \text{ N}$$



b) If ground were smooth, Resultant force's magnitude $\neq 0$

$$c) 60g \cos \theta + 10g \cos \theta = 3R_h \sin \theta$$

$$R_h = \frac{20g \cos \theta + 5g \cos \theta}{\sin \theta} = \frac{(20g \cos \theta + 5g)}{\tan \theta} \quad \left(\frac{1}{\tan \theta} \right)$$

d) As θ increases, $\tan \theta$ increases ($0 < \theta < 90^\circ$) so $\cot \theta$ decreases.

$\therefore R$ decreases, as $20g \cos \theta + 5g$ is constant.

$$e) 550 = \frac{20g \times 1.8 + 5g}{\tan \theta}$$

$$\tan \theta = 0.7305$$

$$\theta = 36.1$$

$$f) 550 = \frac{20g \cos \theta + 5g}{\tan \theta} \quad \tan 45^\circ = 1$$

$$x = \frac{550 - 5g}{20g}$$

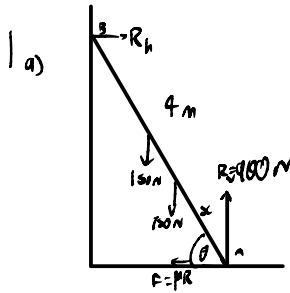
$$= 2.55 \text{ m}$$

Exam Questions

$$\sin \theta = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$$

$$\cos \theta = \frac{1}{\sqrt{10}} = \frac{\sqrt{10}}{10}$$

$$\tan \theta = 3$$



$$R_v = 400 \text{ N}$$

$$R_h = F$$

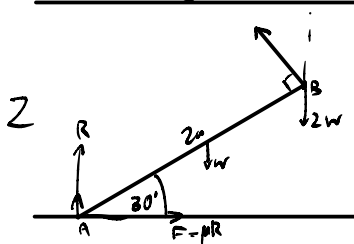
$$\left\{ \begin{array}{l} 750 \times x \cos \theta + 150 \times 2 \cos \theta = F \times 4 \sin \theta \\ \frac{750x}{\sqrt{10}} + \frac{300}{\sqrt{10}} = \frac{12F}{\sqrt{10}} \\ 750x + 300 = 12F \\ F = \frac{750x + 300}{12} \\ = \frac{125}{2}x + 25 \\ = \frac{25}{2}(5x + 2) \\ = \frac{25}{2}(2 + 5x) \end{array} \right.$$

$$b) \frac{25}{2}(2 + 5x) = \frac{400}{4}$$

$$2 + 5x = 18$$

$$5x = 16$$

$$x = \frac{16}{5} = 3.2$$



$$R + B \cos 30 = 3W$$

$$F = B \sin 30$$

$$W \times a \cos 30 + 2W \times 2a \cos 30 = B \sin 30 \times 2a \sin 30 + B \cos 30 \times 2a \cos 30$$

$$\frac{\sqrt{3}Wa}{2} + \frac{4\sqrt{3}Wa}{2} = 2aB$$

$$\frac{B \sin 30}{1} + B \cos 30 = 3W$$

$$5\sqrt{3}Wa = 4aB$$

$$B = \frac{5\sqrt{3}W}{4}$$

$$\frac{5\sqrt{3}}{8\mu} + \frac{15}{8} = 3$$

$$\frac{5\sqrt{3}}{\mu} + 15 = 24$$

$$5\sqrt{3} = 9\mu$$

$$\mu = \frac{5\sqrt{3}}{9}$$

$$6) R^2 + F^2 = \sqrt{3}^2$$

$$R^2 + F^2 = 39$$

$$\downarrow F = MR$$

$$R^2 + M^2 R^2 = 39 \quad R^2(1+M^2) = 39$$

$$R^2 = \frac{39}{1+M^2}$$

$$= \frac{39}{1+\frac{9}{4}} = \frac{81}{4}$$

$$\therefore R = \frac{9}{2}$$

$$\frac{d}{2} = 3W - \frac{5\sqrt{3}}{4} W \times \frac{\sqrt{3}}{2}$$

$$\frac{4}{2} = \frac{9}{8} W$$

$$\underline{W = 4}$$